SAPHIR: Secure Acoustic Pairing for Heterogeneous Inter-agent Rendezvous

Rathod Vaibhav Vishwas Ram
Department of Computer Science and Engineering, R. V. College of Engineering, Bengaluru, India
github.com/vaibhav-rm/SAPHIR

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Abstract

Voice AI agents that encounter each other on a call still converse the wasteful way — speech → text → LLM → speech on both sides — even though both endpoints are machines. The viral Gibberlink demonstration (February 2025) showed two agents switching to data-over-sound tones instead, but its switch is unauthenticated, unnegotiated, unstructured, and opaque to human operators. Meanwhile the emerging interoperability stack (MCP, A2A/ACP, ANP, InterSAGE) uniformly assumes IP connectivity over HTTPS, leaving voice-only calls, PSTN trunks, RF-restricted and offline settings without a pairing story. This paper presents SAPHIR, a minimal authenticated pairing protocol for two heterogeneous agents that share nothing but a bare audio channel: a three-phase handshake (HELLO → WELCOME → KEY1 → KEY2) with ephemeral X25519 keys, compact CBOR task encoding, ChaCha20-Poly1305 authenticated messaging, an optional stop-and-wait ARQ layer for lossy acoustics, and a human-readable sidecar audit log. An open prototype (∼1.6k lines of Python including tests and harnesses) demonstrates a hotel-booking dialogue end to end: 117–139 B task frames across two domains, blocked spoof/replay/downgrade attacks, 100% pairing success on clean channels, 95–100% under bursty VoIP-like loss where raw frames score 0%, and immediate cost breakeven that widens to ∼65× at 20 turns. We state precisely what is measured, what is modeled, and what remains — over-air validation, PKI-backed identity, and group sessions — and close with open research questions.

1 Introduction

In February 2025 a hackathon project called Gibberlink [1] demonstrated two ElevenLabs voice agents on a phone call discovering both sides are AI and switching from English to `ggwave` [2] data-over-sound tones. The video passed 15 M views and the intuition is sound: rendering speech for a machine listener wastes two neural speech passes and thousands of tokens per turn. The implementation, however, is a prompt ("switch if the other side is AI") plus raw tones. Four gaps follow directly: (1) the switch is *unauthenticated* — any speaker can emit beeps; (2) the format is *unnegotiated* — both sides must hardcode the same profile; (3) the content is *unstructured English* — tones carry sentences, not intents; (4) the exchange is *opaque to humans* — operators cannot read what was agreed. A recorded beep replays; a crafted beep injects; a legacy peer is silently beeped at.

In parallel, 2024–26 produced the first serious agent-interoperability stack: Anthropic's MCP for agent-to-tool access, Google's A2A (now Linux Foundation) for task delegation via Agent Cards, IBM/Cisco ACP variants for performative messaging, ANP for decentralized discovery with W3C DIDs, and InterSAGE as the identity/trust substrate [3, 4]. Security analysis is maturing alongside:

A2ASecBench [5] and ASB systematize protocol-logic attacks that bypass default safeguards, and communication-graph analysis [6] shows even encrypted transports leak who-talks-to-whom. Yet every protocol in this stack assumes URL-addressable agents over HTTP(S). IETF framework drafts [8] list discovery, negotiation, and fallback among open requirements. None answers: *what if the only shared channel is sound?* Near-term agent telephony (AI receptionists, AI assistants calling businesses), PSTN trunks that destroy ultrasound, RF-restricted facilities, and offline/disaster settings all produce voice-only encounters between heterogeneous agents.

Contribution. We argue acoustic pairing deserves first-class protocol status and present SAPHIR (Secure Acoustic Pairing for Heterogeneous Inter-agent Rendezvous): detect, negotiate, encrypt, collaborate in structured intents, and account for every exchange in a human-readable sidecar. We contribute (i) the protocol (handshake, compact encoding, AEAD messaging, optional ARQ); (ii) an explicit threat model with verified blocks; (iii) an open prototype with seven evaluation harnesses; and (iv) a disciplined measured-vs-modeled accounting that states exactly what is (and is not) claimed. All artifacts are public to enable red-teaming.

2 Background and related work

Gibberlink and ggwave. Gibberlink [1] contributes the core observation (speech is wasteful machine-to-machine) and a demo, not a protocol: no framing beyond ggwave frames, no handshake, no security. `ggwave` [2] is a data-over-sound modem (FSK + Reed-Solomon, tens of bytes/sec in robust audible profiles) in the lineage of dial-up modems; SAPHIR treats it as a byte pipe and adds everything above it.

The HTTP interop stack. The four-protocol survey [3] (MCP/ACP/A2A/ANP) proposes a phased roadmap from tool access to decentralized agent marketplaces — all over IP. InterSAGE [4] supplies the missing trust substrate (persistent identity, capability-aware discovery, attenuated delegation, audit trails) and explicitly complements, rather than replaces, transport protocols. SAPHIR's claim is narrow and compatible: it is the pairing layer for the one transport the stack omits, and its ephemeral-DID handshake is designed to upgrade to InterSAGE-style credentials where IP exists (hybrid mode).

Attack literature. A2ASecBench [5] taxonomizes supply-chain and protocol-logic attacks across A2A stages; ASB benchmarks prompt-injection, memory poisoning, and backdoors across agent backbones; both find high attack success against default configurations. Our threat model imports their methodology (named attacker capabilities, verified blocks) at the acoustic layer, where the cheapest attacks — replay and injection — need only a speaker.

Semantics and metadata. Two recent critiques shape our design: well-formed JSON does not imply shared meaning [7], motivating our ontology-tagged intent slots over raw English; and encrypted content still exposes the communication graph, whose action-coupled metadata threatens workflow integrity [6], motivating our sidecar's selective disclosure as future work rather than a solved claim.

Deployment context. Industry surveys report the trust gap blocking agentic-AI production (85% exploring, 5% in production) [9]; IETF drafts [8] formalize the missing negotiation/fallback machinery. SAPHIR targets exactly that intersection for voice-only encounters.

3 Problem and threat model

Two heterogeneous voice agents share a low-bandwidth ($\sim$30 Bps reference), possibly lossy, possibly adversary-observable audio channel, plus optionally an IP side channel (hybrid mode). Goals: pair

within seconds, agree on codecs and a task ontology, communicate confidentially with integrity, degrade explicitly (never silently) with legacy peers, and leave a human-inspectable record. Attacker capabilities: observe, record, replay, and emit arbitrary audio; impersonate without keys; force or suppress the switch. Out of scope in this version: attackers holding stolen session keys (needs PKI rotation), traffic-analysis resistance, and denial-of-service on the acoustic medium.

4 The SAPHIR protocol

Framing. Length-prefixed CBOR envelopes, `u32be len` ‖ `CBOR({v,type,seq,sender,nonce,payload})`, cap 200 B (enforced). **Handshake.** HELLO (ephemeral `did:key`, codec/mode lists) → WELCOME (selected codec; peers offering no SAPHIR codec are flagged and kept in English) → KEY1/KEY2 (32 B X25519 ephemeral pubkeys each way; HKDF-SHA256 derives a 32 B ChaCha20-Poly1305 session key; forward secrecy per pairing). **Messaging.** TASK/RESULT carry `ChaCha(cbor({intent,sl`
with the sender as AAD; replays are rejected by sequence plus (nonce, ciphertext) sets. **Compact encoding.** Hotel-ontology shorts (`hotel.book`→`hb`, `date`→`d`, ...) cut task frames 182/202 B → 117–139 B across hotel and taxi domains (non-default ontology tagged, value shorts); legacy full-form still decodes. **ARQ (optional).** Stop-and-wait layer: ≤110 B CRC32 chunks (each fits one audible frame) with ACKs and bounded retries, tick-driven and deterministic. **Audit.** Every frame is appended to a JSONL sidecar with a human summary and decrypted view. Figure 1 summarizes the exchange.

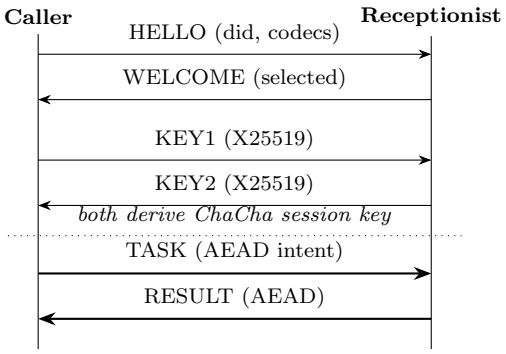


Figure 1: SAPHIR pairing. Negotiation is plaintext; tasks are encrypted; every frame is sidecar-logged.

5 Implementation

The prototype (`github.com/vaibhav-rm/SAPHIR`, MIT) is ∼1.6k lines of Python: `protocol` (envelopes), `session` (handshake state machine), `crypto` (X25519/HKDF/ChaCha), `codec` (compact maps), `channel` (in-memory, lossy, burst-error, ARQ link), `agents` (rule-based hotel and taxi tasks), `audit`, `metrics`. Seven harnesses (`demo`, `bench`, `attacks`, `robustness`, `reliability`, `codec_profiles`, `amortization`) plus 16 pytest tests cover functionality, attacks, and all sweeps. No audio hardware, API keys, or network are required; channels are in-process byte pipes with documented seeds.

6 Evaluation

Functionality. Full pairing plus a two-intent booking dialogue runs end to end; measured sizes: handshake 609 B one-time, HELLO ∼194 B, KEY ∼107–113 B, all four TASK frames 117–139 B compact (hotel and taxi domains), RESULT 126–139 B.

Attacks (Table 1). A keyless attacker is rejected (`InvalidTag`); exact replays are rejected (sequence reuse); legacy peers are kept in English with an explicit flag. All three ship as failing-closed checks in `evals/attacks.py`.

Table 1: Threat outcomes: raw tones vs. SAPHIR (prototype-verified).

Attack	Raw tones	SAPHIR
Tone spoof / injection	succeeds	blocked (AEAD tag)
Replay of valid frame	succeeds	blocked (seq + nonce set)
Forced downgrade	silent	flagged, stays in English
Eavesdropping	plaintext	ciphertext
Stolen session keys	n/a	out of scope (needs PKI)
Who-talks-to-whom	n/a	out of scope

Raw-channel sweep (50 trials/cell, no retransmission): 100% clean, 94% at 1% loss, 44% at 10% loss; corruption cliff (36% at 0.1%, 0% at 0.5%) since AEAD must reject forgeries — integrity over availability, by design. *ARQ sweep (20 trials/cell, Table 2):* the same grid converts the cliff into retry cost: 100%/95% where raw scored 0%, at ∼1.6× wire bytes; residual 10%-loss failures are bounded-budget retry exhaustion.

Table 2: ARQ vs. raw success rate per grid cell (drop, corruption).

Cell	Raw	ARQ	Median retries (ARQ)
(0.00, 0.000)	100%	100%	0
(0.05, 0.000)	70%	100%	1
(0.00, 0.005)	0%	100%	8
(0.05, 0.005)	0%	95%	12
(0.10, 0.000)	44%	85%	3

Codec/noise profiles (byte-level burst models, 20 trials): clean 100%, cafe-noise 95%, bursty VoIP-like 95%, harsh narrowband 80% — bounding the design's operating envelope; waveform/over-air measurement is explicitly future work. *Amortization (Fig. 2):* the 609 B handshake amortizes over turns (mean 263 B wire per pair); modeled neural-cost breakeven is immediate and ∼65× at 20 turns. Wall-clock is deliberately not claimed: narrowband audio moves fewer bytes/sec than speech, so SAPHIR trades time for near-zero neural compute; wire bytes stay above raw-text-over-tones (structure plus auth tags), shown, not hidden.

7 Limitations and open questions

(1) No room-acoustics validation yet — loss models are byte-level, though all framing now round-trips through the real ggwave implementation in-process and a browser demo drives genuine over-air exchange; Opus/G.711 traces and measured rooms are the next experiment. A learned lesson belongs here: the ggwave implementation shows content-dependent first-try decode misses, so retries must re-seal with fresh randomness rather than retransmit identical bytes — our demo does exactly

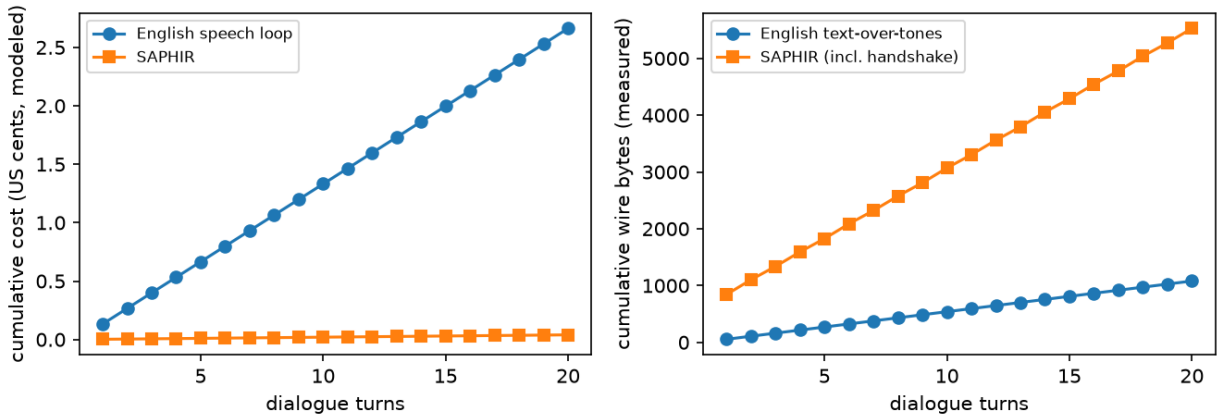


Figure 2: Cumulative modeled cost (left) and measured wire bytes (right) vs. dialogue turns. Cost wins immediately; bytes do not — by design and on the record.

this. (2) Identity is ephemeral, not PKI-backed; upgrading the HELLO to verifiable credentials (where IP exists) is designed but unbuilt. Hand dictionaries are demo-scale; schema-driven dictionaries negotiated in HELLO are the scalable compression story and remain future work. (3) Two-party only; MLS-style groups are future work. (4) Auditing stops at readability; selective disclosure without workflow-graph leakage [6] is open. We invite red-teamers: the artifact reproduces every table from one command each (see `docs/REPRODUCIBILITY.md`).

8 Conclusion

Gibberlink asked the right question — why speak to machines in speech? — and gave a demo answer. SAPHIR gives a protocol answer for the channels the interop stack forgot: negotiate briefly, encrypt cheaply, speak in structure, and keep every beep readable. The evidence here is preliminary but exact about its boundaries; the operating envelope, the cost curves, and the open challenges are published alongside the code so the community can push past them.

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